

# MELODY XUAN LIM, PH.D

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## EDUCATION

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### University of Chicago

PhD in Physics

Advisor: Heinrich M. Jaeger

Thesis: Acoustically Levitated Granular Matter

Chicago, IL

August 2016 – March 2022

### Duke University

BSc in Physics

Summa Cum Laude

Graduation with Distinction

Durham, NC

August 2012 – May 2016

## ACADEMIC EMPLOYMENT

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### Cornell University

Postdoctoral Research Associate

Department of Physics

Laboratory of Atomic and Solid State Physics

Supervisors: Itai Cohen, Paul McEuen

Funded by: KIC Experimental Postdoctoral Fellowship

Ithaca, NY

March 2022 – present

## AWARDS AND HONORS

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### Academic Awards

- Whetten Memorial Award, Cornell Nanoscale Facility 2023  
*In recognition of outstanding young women in science and engineering whose research was conducted in the CNF.*
- KIC Experimental Postdoctoral Fellowship, Cornell University 2022  
*In support of the best and brightest young researchers in nanoscale science to pursue postdoctoral work at Cornell.*
- Yodh Prize, University of Chicago 2021  
*In recognition of outstanding experimental work by a graduate student in physics*
- Soft Matter for All Symposium 2020  
*In promotion of early-career researchers in the field of Soft and Living Matter*
- Eckhardt Scholar, University of Chicago 2016–2022  
*Awarded to support exceptional doctoral students*
- Maria Goeppert-Mayer fellowship, University of Chicago 2016–2017
- Sachs fellowship, University of Chicago 2016–2017
- Phi Beta Kappa member
- Daphne Chang Memorial Award, Duke University 2015  
*Awarded to students who excel in undergraduate research*
- Lemelson-MIT Student Prize Finalist 2016

### Travel & Speaker Awards

- Future Faculty Symposium Travel Award, Society of Engineering Science 2025
- Best poster, New York Nanoscale Symposium 2023
- Best talk, Research Triangle CUWiP 2015
- Poster Award, 1st prize, Duke University undergraduate presentation 2010, 2011

## INVITED TALKS

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|                                                                    |      |
|--------------------------------------------------------------------|------|
| American Physical Society March Meeting                            | 2026 |
| Society of Engineering Science Annual Technical Meeting            | 2025 |
| Hong Kong University of Science and Technology, Physics Department | 2025 |
| Cornell University Soft Matter Seminar                             | 2025 |
| Syracuse University Soft Matter and Biophysics Seminar             | 2025 |
| American Physical Society March Meeting                            | 2022 |
| Cornell University Kavli Institute Special Seminar                 | 2022 |
| Soft Matter for All Symposium                                      | 2020 |
| James Franck Institute Friday Seminar                              | 2020 |
| James Franck Institute Friday Seminar                              | 2019 |
| Granular Matter Gordon Research Conference                         | 2018 |
| James Franck Institute Friday Seminar                              | 2018 |

## TEACHING, OUTREACH, AND SERVICE

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### Mentees

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|-------------------------------------------|-----------|
| · Winter Knutson, undergraduate student   | 2024      |
| · Naomi Naranjo, undergraduate student    | 2023      |
| · Brady Wu, first year graduate student   | 2020–2021 |
| · Annie Xia, undergraduate student        | 2019–2026 |
| · Tali Khain, first year graduate student | 2019      |
| · Tchakamau Ra, postbac researcher        | 2019–2021 |
| · Adam Kline, first year graduate student | 2018–2019 |
| · Isaac Harris, undergraduate student     | 2017–2019 |

### Journal Review

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| · Europhysics Letters, Granular Matter, Applied Physics Letters, Astronomy and Astrophysics, Experimental Astronomy, Sensors and Actuators, Journal of the Acoustical Society of America |  |
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### Outreach

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|------------------------------------------------------------------|-----------|
| · Cornell Nano Day                                               | 2024      |
| · Cornell Expanding Your Horizons, Workshop Leader               | 2024      |
| · University of Chicago annual open house, “Physics with a Bang” | 2016–2019 |

### Service to Professional Societies

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| · Session organizer, Division of Nonlinear and Statistical Physics, American Physical Society | 2026 |
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## LIST OF PUBLICATIONS

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### Peer-reviewed journal articles

- Liang, Z.\*, **Lim, M. X.\***, Zhu, Q., Mottes, F., Kim, J. Z., Guttieres, L., Smart, C., Pearson, T., Du, C. X., Brenner, M. P., McEuen, P. L., & Cohen, I. Magnetic Decoupling for High-Yield, Time-Efficient Microscale Self-Assembly, *Proceedings of the National Academy of Science*, 122 (35) e2502361122. (\* indicates equal contribution).
- C.L. Smart, T. G. Pearson, Z.Liang, **M.X. Lim**, M.I. Abdelrahman, F. Monticone, I. Cohen, P.L. McEuen. Magnetically programmed diffractive robotics , *Science* **386**(6725), 2024.
- B. Wu, B. vanSaders, **M.X. Lim**, H.M. Jaeger. Hydrodynamic coupling melts acoustically levitated crystalline rafts, *Proceedings of the National Academy of Science*, 120 (29) e2301625120.
- **M.X. Lim**, H.M. Jaeger. Acoustically levitated lock and key grains, *Phys. Rev. Research*, **5**, 013116.
- **M.X. Lim**, B. vanSaders, A. Souslov, H.M. Jaeger. Mechanical properties of acoustically levitated granular rafts, *Phys. Rev. X*, **12**, 021017.
- **M.X. Lim**, K.A. Murphy, H.M. Jaeger. Edges control clustering in levitated granular matter. *Granular Matter*, **21**(3), 77, 2019
- **M.X. Lim**, A. Souslov, V. Vitelli, H.M. Jaeger. Cluster formation by acoustic forces and active fluctuations in levitated granular matter. *Nature Physics*, **15**(5), 460–464, 2019
- **M.X. Lim**, R.P. Behringer. Topology of force networks in granular media under impact. *EPL*, **120**(4), 44003, 2018
- **M.X. Lim**, J. Barés, H. Zheng, R.P. Behringer. Force and mass dynamics in non-Newtonian suspensions. *Physical Review Letters* **119**(18), 184501, 2017
- **M.X. Lim**, H.S. Greenside. The external magnetic field created by the superposition of identical parallel finite solenoids. *American Journal of Physics* **84**(8), 606–615, 2016

### Peer-reviewed journal articles authored by mentees

- A.G. Kline, **M.X. Lim**, H.M. Jaeger. Precision measurement of tribocharging in acoustically levitated sub-millimeter grains. *Review of Scientific Instruments*, **91**(2), 023908, 2020
- I.A. Harris, **M.X. Lim**, H.M. Jaeger. Temperature dependence of nylon and PTFE triboelectrification. *Phys. Rev. Materials*, **3**(8), 085603, 2019

### Review Articles

- **M.X. Lim**, B. vanSaders, H.M. Jaeger. Acoustic manipulation of multi-body structures and dynamics, *Reports on Progress in Physics* **87**(6), 2024.

### Patents

- Portable spirometer and method for monitoring lung function G Poore, D Chander, Jinsu Kim, **M.X. Lim** - US Patent 10,098,570, 2018

### In Submission

- **M.X. Lim**, Z. Liang, G. Alkuino, I. Griniasty, T. Zhang, P.L. McEuen, I. Cohen. Multistable magnetoelastic energy landscapes for autonomous microscopic machines.

- A. Xia, **M.X. Lim**, J.Z. Kim, B. vanSaders, H.M. Jaeger. Discovering the emergent nonlinear dynamics of acoustically levitated cube clusters.

## REFERENCES

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Heinrich M. Jaeger

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Paul L. McEuen

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